DIGITAL NOTES • SOFT COMPUTING

UNIT – V

Rough Sets & Integration of Soft Computing

What is inside this unit

1	Rough sets – indiscernibility, lower/upper approximation, dependency
2	Reducts, core and the discernibility matrix
3	Rule induction – certain and possible rules, quality measures
4	Integration of soft computing techniques (neuro-fuzzy, genetic-fuzzy, neuro-genetic, rough hybrids)
5	Solved problem & revision sheet

After this unit you will be able to

- Compute lower and upper approximations and accuracy of a rough set.
- Build a discernibility matrix and find reducts and core.
- Generate IF–THEN rules from a decision table.
- Explain why and how soft-computing techniques are integrated.

How to use these notes

Read the **blue headings** first, study each diagram, then read the points. Yellow boxes = exam tips / key formulas. Green boxes = quick memory tricks. The last page has a one-page revision sheet and likely exam questions.

1 Rough Sets

Rough set theory was proposed by **Zdzisław Pawlak (1982)** to handle **vague and incomplete knowledge** using **only the data itself**. Idea: objects that look the same (same attribute values) cannot be told apart – so a concept can often only be described by an **approximate (rough)** boundary.

Fuzzy sets	Rough sets
Vagueness described by **membership degree μ**	Vagueness described by **lower and upper approximations**
Needs a membership function (from expert)	**No extra information** needed – uses the data only
Handles gradual concepts (“tall”)	Handles indiscernibility and incomplete data
Output: degrees of truth	Output: reductions and decision rules

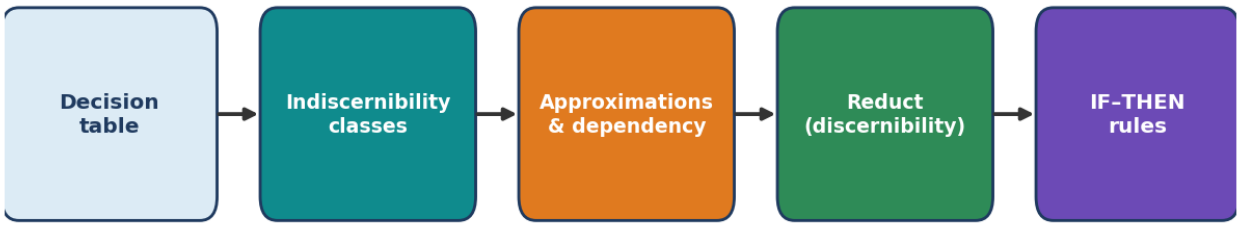


Fig 1.1 Overall rough-set analysis pipeline

1.1 Information system and decision table

Data is kept as a table. Rows = objects (U), columns = **condition attributes C** and one **decision attribute D**. Example – patients and flu:

Object	Headache (H)	Muscle-pain (M)	Temp (T)	Flu (D)
1	yes	yes	normal	**no**
2	yes	yes	high	**yes**
3	yes	yes	very high	**yes**
4	no	yes	normal	**no**
5	no	no	high	**no**
6	no	yes	very high	**no**

1.2 Indiscernibility relation

Two objects are **indiscernible** w.r.t. a set of attributes B if they have the **same values for all attributes in B**. This is an **equivalence relation**; it splits U into **equivalence classes** (granules) written $\mathbf{[x]_B}$.

Example

Take B = {Muscle-pain, Temp}. Objects 1 and 4 both have (yes, normal) → indiscernible. Objects 3 and 6 both have (yes, very high).
U / IND(B) = { {1,4}, {2}, {3,6}, {5} }

1.3 Lower and upper approximation

Let X ⊆ U be the concept we want to describe, e.g. X = patients with Flu = yes. Using the classes of B:

- **Lower approximation $\underline{B}X$** = union of classes **completely inside** X → objects that **certainly** belong to X.
- **Upper approximation $\overline{B}X$** = union of classes that **touch** X (overlap) → objects that **possibly** belong to X.
- **Boundary region $BN_B(X)$** = $\overline{B}X - \underline{B}X$ → cannot be decided. **X is rough if boundary ≠ ∅**, crisp (exact) if boundary = ∅.

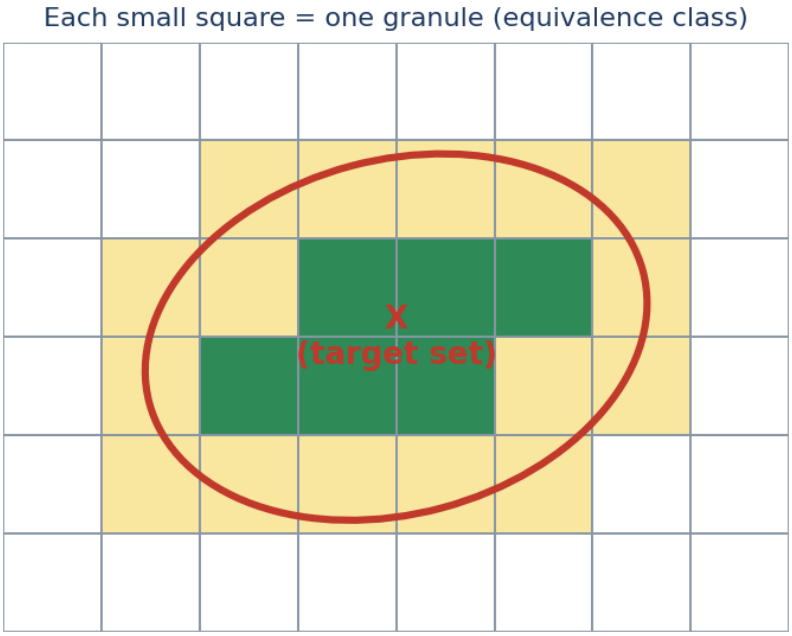


Fig 1.2 Granules of knowledge, the target set X and its approximations

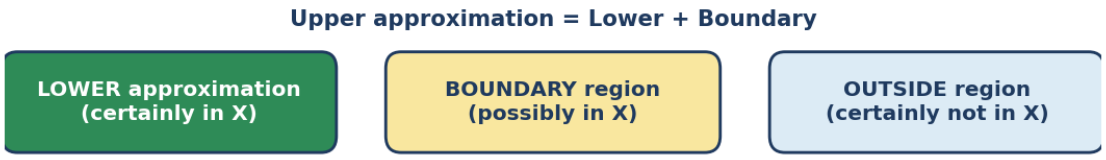


Fig 1.3 Colour key

Worked example – approximations of Flu = yes using B = {M, T}

X = {2,3} (patients 2 and 3 have flu). Classes: {1,4}, {2}, {3,6}, {5}
Lower $\underline{B}X$ = classes fully inside X = **{2}**
Upper $\overline{B}X$ = classes overlapping X = {2,3,6} = **{2,3,6}**
Boundary = {3,6} (objects 3 and 6 look identical on M and T, but 3 has flu and 6 does not!)
Outside = {1,4,5}
Accuracy of approximation α = |lower| / |upper| = 1/3 = 0.33 (1 = crisp, 0 = totally rough)

Formulas to remember

• $\alpha_B(X) = |\underline{B}X| / |\overline{B}X|$ • Roughness = 1 − α • Positive region $POS_B(D)$ = union of lower approximations of all decision classes
• Degree of dependency $\gamma_B(D) = |POS_B(D)| / |U|$ ($\gamma = 1 \rightarrow$ D fully depends on B)

1.4 Dependency of attributes

With B = {M, T}: positive region = {1,2,4,5} → **γ = 4/6 = 0.67** (objects 3 and 6 spoil the dependency). With all attributes C = {H, M, T}: every object is unique, so γ = 6/6 = **1** – the decision depends completely on the conditions.

1.5 Rough membership function

Rough sets can also give a **degree**: $\mu_X^B(x) = |[x]_B \cap X| / |[x]_B|$ – the fraction of the object's class that lies in X. Unlike a fuzzy μ, it is **computed from data**, not supplied by an expert.

Object	Class [x]	$\lvert[x]\cap X\rvert / \lvert[x]\rvert$	μ	Status
1	{1,4}	0/2	0.00	certainly not
2	{2}	1/1	1.00	certain in X
3	{3,6}	1/2	0.50	possible
4	{1,4}	0/2	0.00	certainly not
5	{5}	0/1	0.00	certainly not
6	{3,6}	1/2	0.50	possible

1.6 Extensions of rough sets

Extension	Idea
Variable precision rough set (VPRS)	Allows a small error β – a class counts as "inside" X if ≥ (1 − β) of it is in X (handles noise)
Fuzzy-rough sets	Equivalence classes and the target set are fuzzy – handles continuous data
Tolerance / similarity-based	Uses similarity instead of strict equality of values
Dominance-based (DRSA)	Handles ordered (preference) criteria such as low < medium < high

2 Reduces, Core, Discernibility Matrix

Not all attributes are needed. A **reduct** is a **minimal subset of condition attributes** that gives the **same classification power** (same γ) as the full set. The **core** is the set of attributes that are in **every** reduct – the indispensable ones: Core = ∩ Reducts.

2.1 Discernibility matrix

Build a symmetric matrix of objects. Each cell (i, j) lists **the condition attributes in which objects i and j differ**. For decision-relative reducts, fill only cells where objects have **different decisions** (others are "–").

	1	2	3	4	5	6
1						
2	T					
3	T	–				
4	–	H,T	H,T			
5	–	H,M	H,M,T	–		
6	–	H,T	H	–	–	

Fig 2.1 Decision-relative discernibility matrix for the flu table (lower triangle)

2.2 Discernibility function and reduct

- Write each non-empty cell as a **sum (OR)** of its attributes; combine all cells with **AND** (product).
- Simplify with Boolean laws (absorption: a · (a + b) = a). Each **prime implicant** = one **reduct**.

Worked example

Cells found: (H), (T), (H + M), (H + T), (H + M + T)
f(H, M, T) = T · (H + T) · (H + M) · (H + M + T) · H
Absorption: T · (H + T) = T, and H · (H + M) = H, etc. ⇒ **f = H · T**
Reduct = {H, T} = {Headache, Temp}, Core = {H, T}. **Muscle-pain (M) is redundant** and can be dropped without losing any classification power.

Exam tip – steps

1) Find all differing-attribute cells for object pairs with different decisions. 2) Write the CNF. 3) Simplify by absorption. 4) Prime implicants = reducts; single-attribute cells give the core.

2.3 QuickReduct – dependency-based search (worked)

Dependency of the decision on single attributes: γ(H) = 0.50, γ(M) = 0.17, γ(T) = 0.33. Take the best single attribute, then add the one that raises γ most:

Subset	γ	Comment
HM	0.50	
HT	1.00	γ = 1 → reduct found
MT	0.67	

The pair {H, T} reaches γ = 1 with only two attributes, which agrees with the discernibility-matrix result.

2.4 Other reduct methods

- **Dependency-based (QuickReduct):** start empty, add the attribute that increases γ most, stop when γ equals that of the full set.
- **Heuristic / evolutionary:** GA or PSO search subsets when the number of attributes is large (finding the shortest reduct is NP-hard).

3 Rule Induction

Rule induction converts the (reduced) decision table into **IF–THEN rules** that explain and predict the decision. Rough sets give rules that are **compact, readable and data-driven**.

3.1 Steps

- **1.** Compute a **reduct** and remove redundant attributes (here keep H, T).
- **2.** Write one rule per object from the reduced table: IF (attribute = value) AND … THEN decision.
- **3. Merge / simplify**: drop conditions that are not needed (value reducts) and remove duplicate rules.
- **4.** Measure rule quality: support, confidence, coverage.

Rule	IF … THEN …	Objects	Support	Confidence
R1	IF Headache = no THEN Flu = no	4, 5, 6	3	100 %
R2	IF Headache = yes AND Temp = normal THEN Flu = no	1	1	100 %
R3	IF Headache = yes AND Temp = high THEN Flu = yes	2	1	100 %
R4	IF Headache = yes AND Temp = very high THEN Flu = yes	3	1	100 %

Rules R3 and R4 may be merged: **IF Headache = yes AND Temp ∈ {high, very high} THEN Flu = yes** (support 2). Four short rules replace the whole six-row table, and the

attribute Muscle-pain is no longer used.

3.2 Certain and possible rules

Type	Made from	Confidence	Example (using B = {M, T})
Certain (exact) rule	Lower approximation	100 %	IF Muscle-pain = yes AND Temp = high THEN Flu = yes (object 2)
Possible (approximate) rule	Boundary / upper approximation	< 100 %	IF Muscle-pain = yes AND Temp = very high THEN Flu = yes (50 % – objects 3, 6)

3.3 Rule quality measures

Measure	Formula	Meaning
Support	|objects matching IF and THEN|	How many objects back the rule
Confidence (accuracy)	|IF ∧ THEN| / |IF|	Reliability of the rule
Coverage	|IF ∧ THEN| / |objects of that decision|	How much of the class is explained

Example R1: IF H = no (objects 4,5,6) THEN Flu = no. Support = 3, confidence = 3/3 = 100 %, coverage = 3/4 = 75 % (4 patients have no flu).

3.4 Applications of rough sets

- **Feature selection / dimensionality reduction** before neural networks or fuzzy systems.
- Medical diagnosis, risk analysis, bankruptcy and credit prediction.
- Data mining, text classification, web usage mining, image segmentation, fault diagnosis.
- Decision support where **explainable rules** are required.

3.5 Advantages and limitations

Advantages	Limitations
Needs no prior information (no μ, no probability)	Works on **discrete** data – continuous attributes must be discretised
Finds minimal attribute sets and readable rules	Sensitive to noise; strict indiscernibility
Simple, easy to understand mathematically	Reduct finding is NP-hard for large tables

4 Integration of Soft Computing Techniques

Each soft-computing tool has weaknesses. **Integration (hybridisation)** combines two or more tools so that **one's strength covers another's weakness**. This is Zadeh's vision: SC is a **partnership**, not a competition.

Tool	Strength	Weakness
Fuzzy logic	Human-like knowledge, handles vagueness	Cannot learn; rules designed by hand
Neural network	Learns from data, generalises	Black box; needs lots of data
Genetic algorithm / PSO	Global optimisation	Slow; no knowledge representation
Rough sets	Reducts and rules from raw data	Needs discrete data

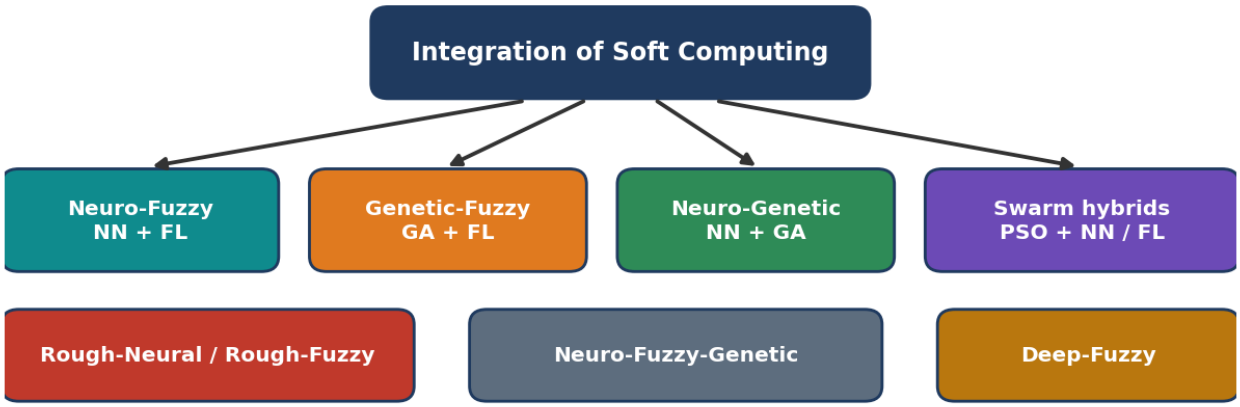


Fig 4.1 Families of hybrid soft-computing systems

4.1 Neuro-fuzzy systems (NN + FL)

- A neural network **learns** the membership functions and rules of a fuzzy system → **learning + interpretability**.
- Best known: **ANFIS** (Adaptive Neuro-Fuzzy Inference System, Jang 1993) – a Sugeno fuzzy system implemented as a 5-layer network.

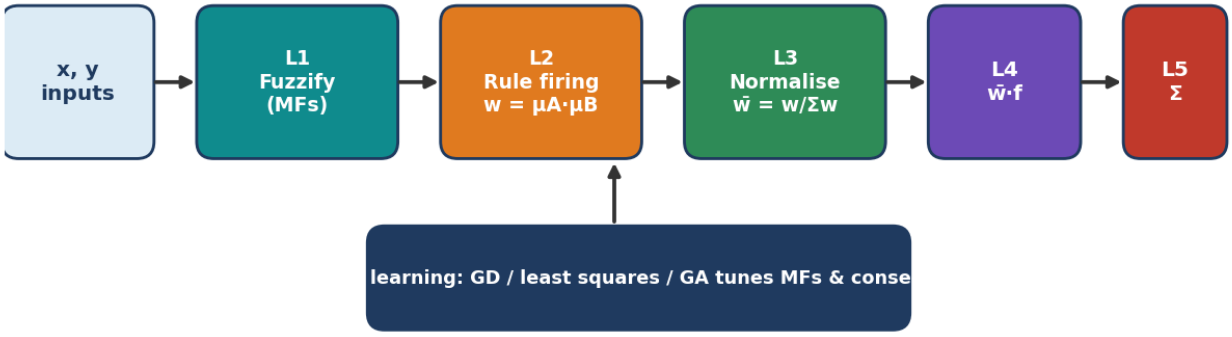


Fig 4.2 ANFIS architecture (5 layers)

Layer	Function
1 – Fuzzification	Node outputs μ(x) of input to a fuzzy set (premise parameters adapt)
2 – Rule layer	Firing strength w = AND of memberships (product)
3 – Normalisation	$\bar{w} = w / \Sigma w$
4 – Consequent	$\bar{w} \cdot f$, where $f = p \cdot x + q \cdot y + r$ (consequent parameters adapt)
5 – Output	$\Sigma \bar{w} \cdot f \rightarrow$ crisp output

Worked example – one ANFIS (Sugeno) forward pass

Rules: R1: IF x is A_1 THEN $f_1 = 2x + 1$ R2: IF x is A_2 THEN $f_2 = x + 4$. Input x = 3, with $\mu_{A_1}(3) = 0.6$, $\mu_{A_2}(3) = 0.2$.
Layer 2: $w_1 = 0.6$, $w_2 = 0.2$. Layer 3: $\bar{w}_1 = 0.6/0.8 = 0.75$, $\bar{w}_2 = 0.25$.
Layer 4: $f_1 = 2 \cdot 3 + 1 = 7$, $f_2 = 3 + 4 = 7 \rightarrow \bar{w}_1 f_1 = 5.25$, $\bar{w}_2 f_2 = 1.75$.
Layer 5 output = 5.25 + 1.75 = 7.0. During training, the parameters of A_1, A_2 and the coefficients of f_1, f_2 are adjusted to reduce error.

4.2 Genetic-fuzzy and swarm-fuzzy systems (GA/PSO + FL)

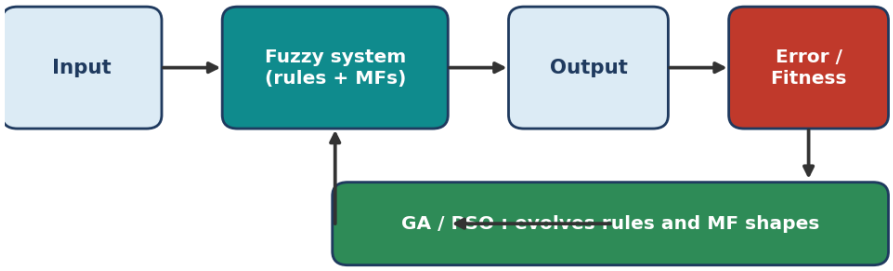


Fig 4.3 GA or PSO automatically designs the fuzzy controller

- The chromosome / particle encodes **MF shapes, number of rules** or rule consequents.
- Fitness = control error or classification accuracy; the best fuzzy system evolves automatically.
- Removes the long manual trial-and-error tuning of fuzzy controllers.

4.3 Neuro-genetic systems (NN + GA)

- GA **trains weights** (avoids local minima of back-propagation) or **chooses the network architecture** (layers, neurons, learning rate).
- Modern form: neuro-evolution and Neural Architecture Search.

4.4 Rough-neural and rough-fuzzy systems

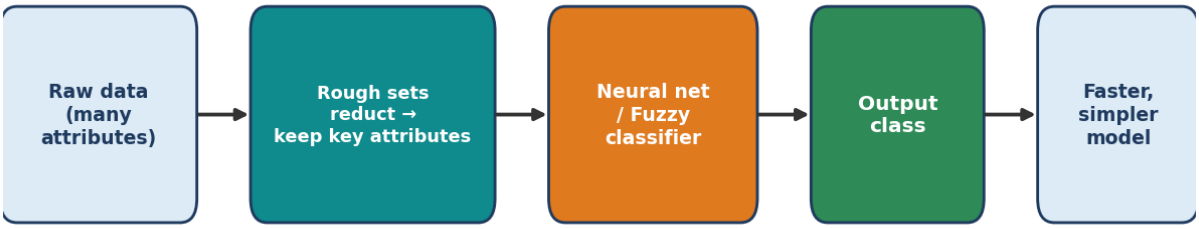


Fig 4.4 Rough sets reduce the attributes before a neural / fuzzy model is built

- **Rough sets** remove irrelevant attributes → smaller, faster, more accurate neural/fuzzy model.
- **Rough-fuzzy:** fuzzy granules + rough approximations handle both vagueness and indiscernibility.
- Rules from rough sets can be used to **initialise** a fuzzy or neural system.

4.5 Ways of combining (coupling)

Type	How	Example
Sequential (loose)	Output of one tool is input of the next	Rough-set reduct → neural classifier
Auxiliary	One tool helps tune another	GA tunes fuzzy MFs
Embedded (tight)	Tools share one structure	ANFIS (fuzzy inside a network)
Fusion / parallel	Tools run side by side and outputs combined	Ensemble of NN + fuzzy classifiers

Hybrid	Typical application
Neuro-fuzzy (ANFIS)	Time-series prediction, adaptive control, medical diagnosis
Genetic-fuzzy	Automatic controller design, rule-base learning, scheduling
Neuro-genetic	Weight optimisation, architecture search, stock prediction
Rough-neural / rough-fuzzy	Feature selection, data mining, gene-expression analysis
Swarm-neural	Training networks with PSO, power-system forecasting

Remember

Fuzzy + NN → learn and explain. GA/PSO + Fuzzy or NN → tune and design. Rough sets + others → clean data and extract rules.

5 Solved Problem & Revision

Problem – rough set analysis

Universe U = {a, b, c, d, e}. Using attribute set B the classes are {a, b}, {c}, {d, e}. Let X = {a, c, d}.
Lower = classes fully inside X = {c}. **Upper** = classes touching X = {a, b} ∪ {c} ∪ {d, e} = {a, b, c, d, e}.
Boundary = {a, b, d, e}. **Accuracy** = 1/5 = 0.2. X is a **rough set** (boundary not empty).

5.1 Full worked problem – buying a used car

Car	Price (P)	Mileage (M)	Colour (C)	Buy?
1	low	low	red	**yes**
2	low	high	red	**no**
3	high	low	blue	**yes**
4	high	high	blue	**no**
5	low	low	blue	**yes**
6	high	high	red	**no**
7	low	high	blue	**no**

Step 1 – approximations of X = Buy-yes = {1,3,5} using a single attribute

Attribute used	Classes	Lower	Upper	Accuracy
P = Price	{1,2,5,7}, {3,4,6}	∅	{1,2,3,4,5,6,7}	0/7 = 0.00
M = Mileage	{1,3,5}, {2,4,6,7}	{1,3,5}	{1,3,5}	3/3 = 1.00
C = Colour	{1,2,6}, {3,4,5,7}	∅	{1,2,3,4,5,6,7}	0/7 = 0.00

Mileage gives lower = upper = X, so X is **crisp** (accuracy 1): Mileage alone explains the decision. Price and Colour leave a large boundary.

Step 2 – discernibility: cells found: (M), (M+C), (P+M), (P+M+C). Single-attribute cell (M) → **core = {M}**; discernibility function f = M, so the **reduct = {M}**.

Step 3 – rules: R1: IF Mileage = low THEN Buy = yes (support 3, confidence 100 %). R2: IF Mileage = high THEN Buy = no (support 4, confidence 100 %).

Conclusion

Three attributes were reduced to one. Two rules explain all seven cars – a small, readable model found from data alone.

Problem – discernibility function

Suppose the discernibility cells are (a + b), (a + c) and (b + c). Then f = (a + b)(a + c)(b + c) = **ab + ac + bc**.
There are **three reducts**: {a, b}, {a, c}, {b, c}. No single attribute appears in every cell, so the **core is empty**.

Revision sheet

• Rough set (Pawlak 1982): approximate a concept by lower and upper approximation from data only.
• α = |lower| / |upper|; γ = |POS| / |U|. • Boundary empty → crisp set; else rough.
• Reduct = minimal attribute subset with same γ; core = common to all reducts. • Discernibility matrix → CNF → absorption → reducts.
• Rules: certain (lower), possible (boundary); quality = support, confidence, coverage.
• Hybrids: neuro-fuzzy (ANFIS), genetic-fuzzy, neuro-genetic, rough-neural, swarm-based.

Quick self-check

Q1. Who proposed rough sets? → **Z. Pawlak, 1982**
Q2. What is in the boundary region? → **Objects that cannot be classified with certainty**
Q3. What does the core contain? → **Attributes common to all reducts**
Q4. What does ANFIS combine? → **Fuzzy inference + neural learning**

Likely exam questions

- Explain rough sets; define lower, upper approximation and boundary with an example.
- What is a reduct and core? Find them using a discernibility matrix for the given table.
- Explain rule induction in rough sets; give certain and possible rules.
- Explain the integration of soft computing techniques; describe neuro-fuzzy (ANFIS) and genetic-fuzzy systems.
- Compare rough sets and fuzzy sets; list applications.